

Shifted-Beam Array Coil for Highly Focal Transcranial Magnetic Stimulation

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Abstract—This paper presents a novel system for transcranial magnetic stimulation (TMS) using an array of vertically arranged, counter-propagating coils to achieve extremely high focality. The proposed design significantly reduces unwanted field spread and the activated brain volume, creating an E-field peak with a full width at half maximum (FWHM) of approximately 1 cm on the cortical surface. The focality is superior compared to all established designs, and in particular all physically realizable figure-8 coils. Such high level of focusing is a notable improvement over current designs and enables even more precise stimulation of the cerebral cortex.

Index Terms—Transcranial magnetic stimulation, field focusing, focality, coil design, optimization, induced electric field, brain stimulation

I. INTRODUCTION

Transcranial magnetic stimulation (TMS) is a non-invasive method of brain stimulation used extensively for medical research and treatment of psychiatric disorders, in which electromagnetic induction is used to generate electric currents within the brain. The goal of TMS is to achieve a focal E field of some threshold E_{thr} at target depth d within the brain. Due to the high permittivity of brain tissue at low frequencies, electromagnetic fields are strongly attenuated resulting in an unavoidable tradeoff between depth and focality.

The figure-8 coil arrangement proposed in 1988 by [1] remains the most widely used coil, and is considered close to the fundamental limit for TMS coil performance by some metrics [2]. Optimization techniques can be used to generate viable coil geometries *ab initio* [2], but this is limited by the choice of constraints as well as the vast parameter space. Nevertheless, substantial improvements can still be achieved in focality and minimizing activation of non-target regions.

The coil design presented in this paper uses a system of vertically arranged coils to form a shifted-beam array, applying the focusing technique described [3]. This allows a sharper E field than is possible with typical TMS coils positioned tangentially to the head surface. In particular, it is superior in focality compared to realizable figure-8 coils of any size. Although the energy requirements of the system are increased, the necessary currents and coil size are within the capabilities of existing TMS systems.

II. STIMULATION AND PERFORMANCE METRICS

A two layer spherical head model of overall radius 9 cm is used in the simulations presented. The outer 1 cm represents the skull, and the interior the brain; such a model's simplicity allows results to be better compared with existing designs from literature [4]. The system is modelled for 2.9 kHz with $\epsilon_{r,brain} = 6.9 \times 10^4$ and $\epsilon_{r,skull} = 1.28 \times 10^3$. Biological tissue, being non-magnetic, has relative permeability $\mu_r = 1$. All simulations were performed on COMSOL Multiphysics 5.5 using the AC/DC module (magnetic fields interface).

The activated volume V_{active} is calculated as the brain domain (skull domain excluded) with $E > E_{thr} = 40$ V/m, a lower bound for which neuronal activation in the motor cortex has been demonstrated [5]. The TMS motor threshold (MT) varies between individuals, but can be reasonably placed at 100 V/m; TMS safety procedures typically require that induced E fields do not exceed 130% of the MT in the brain and 500 V/m globally, which is obeyed by the presented design [6].

Maximum penetration depth d_{max} is also computed in the full skull/brain model. However, in the case of most TMS applications stimulating sensory nerves, 95% of the relevant neurons are located in the outer 1-5 mm layer of the brain, the cerebral cortex. Efforts at field focusing are therefore of greatest importance in this region, which is responsible for perception and most of the brain's information processing. Various metrics for characterising the performance of TMS coils exist; the half-spread $S_{1/2}$ has been proposed as a universal metric, and is here computed for reference despite not fully capturing all aspects of focusing quality [4].

III. BENCHMARK DESIGN: FIGURE-8 COIL

The figure-8 coil, based on the Magstim 70 mm (diameter) double coil (fig. 1), is the most established type used for focal TMS and is commonly used as a benchmark for assessing new coil geometries. In the simulations performed, the 70 mm figure-8 achieves $V_{active} = 162.76 \text{ cm}^3$, $d_{max} = 3.1 \text{ cm}$, $S_{1/2} = 10.35 \text{ cm}^2$.

By reducing the diameter of the figure-8, the focality can be steadily improved. Extremely low-diameter figure-8 coils have been proposed as being the most ideal in terms of focality [7]; for instance, a 2 cm diameter figure-8 coil theoretically produces a hotspot on the cortex with 1 cm FWHM. However, such small loops are highly reactive and inefficient, requiring

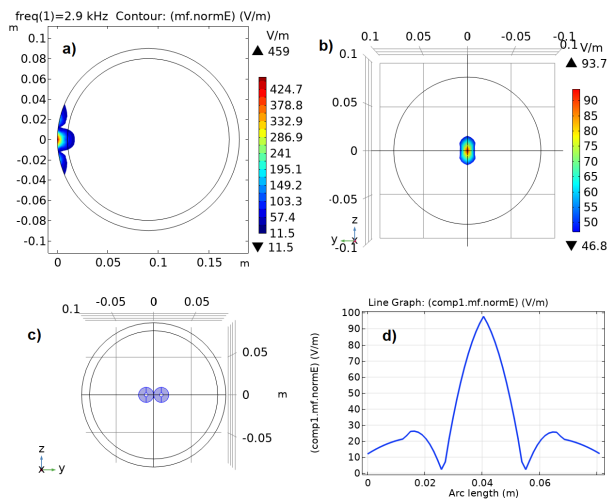


Fig. 1. Extremely small figure-8 coil with 2.0 cm OD, 0.5 cm ID, to demonstrate maximum attainable focality. Coil thickness is 1.1 cm. Each coil is excited with a 18 kA current (an infeasible amount in practice). All coils have $N = 5$ turns. (a) Supra-threshold regions (>40 V/m) for the cortical cut of the model. (b) E field on cortex surface, FWHM 7.3 mm along y axis. (c) View of coil geometry. (d) E field cut line for cortex surface.

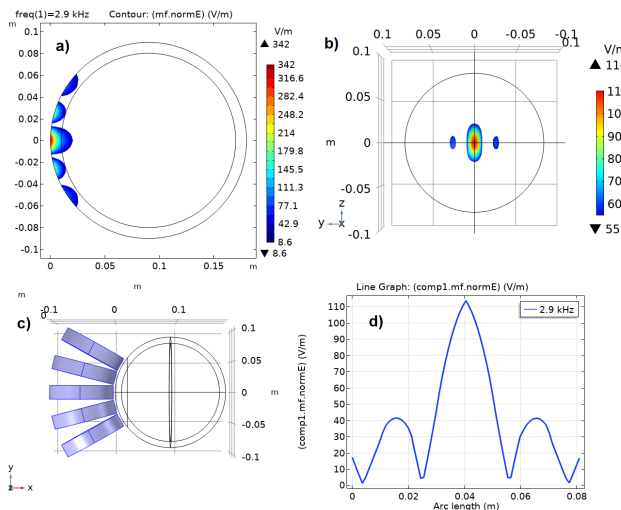


Fig. 2. Shifted-beam array with 7.0 cm outer diameter (OD), 3.0 cm inner diameter (ID) coils. Coil thickness is 2.2 cm. Coil currents are [4824, -10616, 15344, -10616, 4824] A. All coils have $N = 25$ turns. (a) Supra-threshold regions (>40 V/m) for the cortical cut of the model. (b) E field on cortex surface, FWHM 8.8 mm along y axis. (c) View of coil geometry. (d) E field cut line for cortex surface.

extremely high driving currents to produce adequate stimulation effects in the human brain. Furthermore, their small size limits the number of turns that can be incorporated in the coil. In the case of the 2 cm diameter coil this current is easily > 30 kA, producing high Lorentz forces and overheating issues which damages the device after few uses [7]. Therefore it is impractical to consider figure-8 coils of such size, which are so far suited only for use on small animals (mice) using low currents. In the simulations performed, the 2 cm figure-8

coil achieves $V_{active} = 19.58 \text{ cm}^3$, $d_{max} = 1.64 \text{ cm}$, $S_{1/2} = 1.16 \text{ cm}^2$ (fig. 1).

IV. PROPOSED DESIGN: 5X1 SHIFTED-BEAM ARRAY

Using a set of spatially shifted beams as a basis function [3], a focused spot can be generated on the cortex. Due to the plane of the coils being perpendicular (fig. 2) to the surface of the head, the contact area and therefore the focal spot size is reduced. Conforming the vertical array to the surface geometry is beneficial for preserving the focusing; an additional pair of counter-propagating coils is used to control the field spread in the axis perpendicular to the focal axis.

The FWHM of the hotpost is $\sim 0.9 \text{ cm}$ on the cortex, similar to the small figure-8. However, the current required ($\sim 15 \text{ kA}$ for the central element) is readily attainable with current TMS hardware. In the simulations performed, the shifted-beam array achieves $V_{active} = 48.965 \text{ cm}^3$, $d_{max} = 2.0 \text{ cm}$, $S_{1/2} = 3.52 \text{ cm}^2$ (fig. 2).

Using COMSOL's SNOPT optimizer, the coil weights can be adjusted to achieve potentially more superior focusing. A 2D Gaussian beam function ($\sigma = 1 \text{ mm}$) can be specified at the target depth, with the squared difference between the E field taken as the objective to be minimized. The result differs little from the analytic result using the method in [3], suggesting that it is a highly optimal approach to the focusing problem in consideration.

V. CONCLUSION

A multichannel shifted-beam array TMS coil is proposed with substantial improvements in focality and minimization of field spread compared to established designs. Continued optimization of the input currents, coil element geometry and positioning can produce even further improvements. The new coil and methodology used are a notable design departure from conventional TMS coils, presenting new opportunities for continued improvement. This enables more accurate mapping of the cortical surface, and even stimulation of individual gyrii.

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